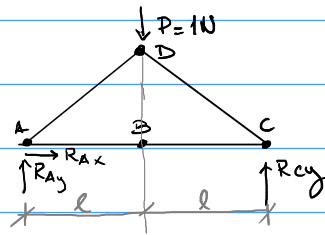


Ej 1:

a)

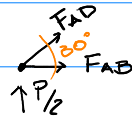
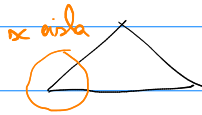


$$\sum F_x = 0 \Rightarrow R_{Ax} = 0$$

$$\sum F_y = 0 \Rightarrow R_{Ay} + R_{Cy} - P = 0 \Rightarrow R_{Ay} = P - R_{Cy} = \frac{P}{2} \quad \left. \begin{array}{l} \approx 0,5 \text{ N} \\ \approx 0,5 \text{ N} \end{array} \right\}$$

$$\sum M_A = 0 \Rightarrow P \cdot l - R_{Cy} \cdot 2l = 0 \Rightarrow R_{Cy} = \frac{P}{2}$$

b)



$$\sum F_x = 0 \Rightarrow F_{AB} + F_{AD} \cdot \cos 30^\circ = 0 \Rightarrow F_{AB} = -F_{AD} \cdot \cos 30^\circ = \frac{P}{2} \cot(30^\circ)$$

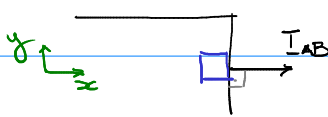
$$\sum F_y = 0 \Rightarrow P/2 + F_{AD} \cdot \sin 30^\circ = 0 \Rightarrow F_{AD} = \frac{-P}{\sin 30^\circ \cdot 2}$$

$$F_{AB} = \frac{P}{2} \cot(30^\circ) = \frac{P \cdot \sqrt{3}}{2} \quad \left. \begin{array}{l} \approx 0,866 \text{ N} \end{array} \right\}$$

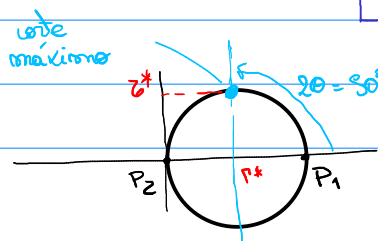
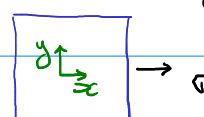
c) Considere el campo de tensiones uniforme y sabiendo que los esfuerzos tienen la dirección del eje longitudinal de la barra. En el caso de la barra A-B, el eje longitudinal tiene la dirección del eje x:

$$T_{AB} = \frac{F_{AB} \cdot e_x}{a_{AB}} = \frac{P \frac{\sqrt{3}}{2} e_x}{\frac{1}{2}} = P \sqrt{3} e_x \quad \left. \begin{array}{l} \approx 1,732 e_x \\ \text{área transversal de la barra A-B} \end{array} \right\}$$

Vector tensión de la barra A-B

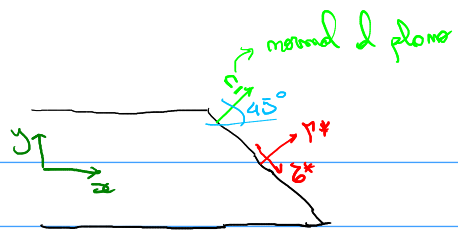


Cuadrado diferencial



$$P_1(7,0)$$

$$P_2(0,0)$$



Ej2:

a)

$$(1) \underline{a} \cdot \underline{c} = 0 \Rightarrow a_i \cdot c_i = 0 \Rightarrow$$

$$\Rightarrow \frac{\partial}{\partial x_j} (a_i \cdot c_i) = \frac{\partial a_i}{\partial x_j} \cdot c_i + a_i \cdot \frac{\partial c_i}{\partial x_j} = 0 \Rightarrow a_i \frac{\partial c_i}{\partial x_j} = -c_i \frac{\partial a_i}{\partial x_j} \Rightarrow$$

$$\Rightarrow \underline{a} \cdot \nabla \underline{c} = -\underline{c} \cdot \nabla \underline{a} \quad (2)$$

$$\nabla \cdot [\underline{a} \times (\underline{b} \times \underline{c})] = \frac{\partial}{\partial x_i} [\underline{a} \times (\underline{b} \times \underline{c})]_i = \frac{\partial}{\partial x_i} [e_{ijk} a_j (\underline{b} \times \underline{c})_k] =$$

$$= \frac{\partial}{\partial x_i} [e_{ijk} a_j e_{kmn} b_m c_n] = e_{kij} e_{kmn} \frac{\partial}{\partial x_i} (a_j b_m c_n)$$

$$= (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) \cdot \left(\frac{\partial a_j}{\partial x_i} b_m c_n + a_j \frac{\partial b_m}{\partial x_i} c_n + a_j b_m \frac{\partial c_n}{\partial x_i} \right)$$

$$= \frac{\partial a_j}{\partial x_i} b_i c_j - \frac{\partial a_j}{\partial x_i} b_j c_i + a_j \frac{\partial b_i}{\partial x_i} c_j - a_j \frac{\partial b_j}{\partial x_i} c_i + a_j b_i \frac{\partial c_j}{\partial x_i} - a_j b_j \frac{\partial c_i}{\partial x_i}$$

$$= \cancel{c \cdot (\nabla \underline{a}) \underline{b}} - \underline{b} \cdot (\nabla \underline{a}) \underline{c} + \cancel{(\underline{a} \cdot \nabla) \underline{c}} - \underline{a} \cdot (\nabla \underline{b}) \underline{c} + \cancel{\underline{a} (\nabla \underline{c}) \underline{b}} - \cancel{(\underline{a} \cdot \nabla) \underline{c}} \quad \begin{matrix} \text{pa(1)} & \text{pa(2)} \end{matrix}$$

$$= -\underline{b} \cdot [(\nabla \underline{a}) \underline{c} + (\underline{a} \cdot \nabla) \underline{c}] - \underline{a} \cdot (\nabla \underline{b}) \underline{c}$$

Solución Alternativa:

$$\nabla \cdot [\underline{a} \times (\underline{b} \times \underline{c})] = \frac{\partial}{\partial x_i} [\underline{a} \times (\underline{b} \times \underline{c})]_i = \frac{\partial}{\partial x_i} [e_{ijk} a_j (\underline{b} \times \underline{c})_k] =$$

$$= \frac{\partial}{\partial x_i} [e_{ijk} a_j e_{kmn} b_m c_n] = \frac{\partial}{\partial x_i} (e_{kij} e_{kmn} a_j b_m c_n) =$$

$$= \frac{\partial}{\partial x_i} \left[(\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm}) a_j b_m c_n \right] = \frac{\partial}{\partial x_i} (a_j b_i c_j - a_j b_j c_i) \quad \text{pa(1)}$$

$$= -\frac{\partial a_j}{\partial x_i} b_j c_i - a_j \frac{\partial b_j}{\partial x_i} c_i - a_j b_j \frac{\partial c_i}{\partial x_i} =$$

$$= -\underline{b} \cdot (\underline{\nabla} \underline{a}) \underline{c} - \underline{a} \cdot (\underline{\nabla} \underline{b}) \underline{c} - (\underline{a} \cdot \underline{b}) \underline{\nabla} \cdot \underline{c}$$

b)

$$e_{ijk} \delta_{mk} \sigma_{jm} = e_{ijk} \tau_{jk} = \frac{1}{2} (e_{ijk} \tau_{jk} - e_{ikj} \tau_{kj}) =$$

por definición de e_{ijk}
 por simetría de τ

$$= \frac{1}{2} (e_{ijk} \tau_{jk} - e_{ijk} \tau_{jk}) = 0$$

g3

a) $\frac{\partial \Pi_i}{\partial x_j} = -b_i$

$$\frac{\partial \Pi_{11}}{\partial x_1} + \frac{\partial \Pi_{12}}{\partial x_2} + \frac{\partial \Pi_{13}}{\partial x_3} = b_1$$

$$2x_1x_2 - 2x_2x_1 + 0 = b_1 \Rightarrow b_1 = 0$$

$$\frac{\partial \Pi_{21}}{\partial x_1} + \frac{\partial \Pi_{22}}{\partial x_2} + \frac{\partial \Pi_{23}}{\partial x_3} = b_2$$

$$1 - x_2^2 + \frac{3x_2^2}{3} - \frac{3}{3} + 0 = b_2 \Rightarrow b_2 = 0$$

$$\frac{\partial \Pi_{31}}{\partial x_1} + \frac{\partial \Pi_{32}}{\partial x_2} + \frac{\partial \Pi_{33}}{\partial x_3} = b_3 \Rightarrow b_3 = 0$$

b)

$$\Pi_{xx} = a^2 \cdot 0 = 0$$

$$\Pi_{yy} = \frac{0^3 - 3 \cdot 0}{3} = 0$$

$$\Pi_{xy} = (1 - 0^2)a = a$$

$$\underline{\underline{\Pi}} = \begin{bmatrix} 0 & a & 0 \\ a & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Pi_{\max/\min} = \frac{0+0}{2} \pm \sqrt{\left(\frac{0-0}{2}\right)^2 + a^2} = \pm a \Rightarrow \begin{cases} \Pi_{\max} = +a \\ \Pi_{\min} = -a \end{cases}$$

c)

$$b_{\max} = \frac{a - (-a)}{2} = a$$

d) Ebene $x_1 + 3x_2 + x_3 = \sqrt{11}(1a+2) \Rightarrow$

$$\underline{n}^* = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \Rightarrow \underline{n} = \frac{1}{\sqrt{1^2+3^2+1^2}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$\underline{I} = \underline{n}^T \cdot \underline{n} = \begin{bmatrix} 0 & a & 0 \\ a & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \cdot \frac{1}{\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{11}} \begin{bmatrix} 3a \\ a \\ 0 \end{bmatrix} = \frac{a}{\sqrt{11}} \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$$